

programme. It has had huge growth mainly due to its geo-navigation map-based search.

ParkingRhino was the official parking partner to ensure seamless

parking operation across 95 parking locations during Kumbhmela 2019 event, held in Prayagraj, Uttar Pradesh. The company is working for several smart city projects in

India. Having extended its system to Philippines already, it is now working towards further expansion and improving the solution through AI.

—Ayushee Sharma

2 NEW TECH MAKES DRONE FLIGHTS STABLE AND SAFER

To operate in space, safety is a paramount challenge when it comes to drones. Drone designing is creating new markets and jobs, and changing social infrastructure and airspace policies.

A drone has two parts: cargo (luggage, camera, etc) and flight (motor, propeller, arm, etc) mechanism. Drones used for logistics, industrial and services are expected to see widespread adoption and use in the future. For example, an inspection drone can attach a camera to the end of a long rod that can enter a small space for inspection. Whereas, a service drone can spray liquids for fumigation or over crops while flying.

Aeronext, a tech startup seeks to overcome the safety issues and collaborate with industries to revolutionise the sky. Aeronext's centre of gravity control, called 4D GRAVITY technology, is for fundamental re-examination and development of the structure of

the unmanned aerial vehicle (UAV).

4D GRAVITY technology optimises the position of the centre of gravity based on flight parameters of the drone coordinated with its movement. It enables a drone to be utilised in an increased number of industrial applications. It is characterised by a separate independent structure that physically separates flight and cargo parts within the drone and gimbals them to displace each other independently. The main theme in drone architecture that Aeronext focuses on is its control by fundamentally redefining the structure of the airframe in the drone.

When a drone is in tilted position, conventional airframe moves its centre of gravity, so some motors experience loads more than the others. Since an Aeronext 4D GRAVITY-equipped aircraft has separate flight and cargo structures, centre of gravity does not shift when tilted. It stabi-

lises the motors and aircraft, as well as improves energy efficiency. As a result, it solves the problems of wind resistance, flight time, reliability and flight speed for drones.

A 4D GRAVITY-powered aircraft improves fuel efficiency by equalising the load on the motors during flight. Not only can the drone perform multipurpose operations, it can also change batteries in flight.

A 4D GRAVITY-powered aircraft is more resistant to changes in the centre of gravity, side winds and updrafts in flight that cause crashes, thus providing a more stable flight. This improves safety and increases flight speed.

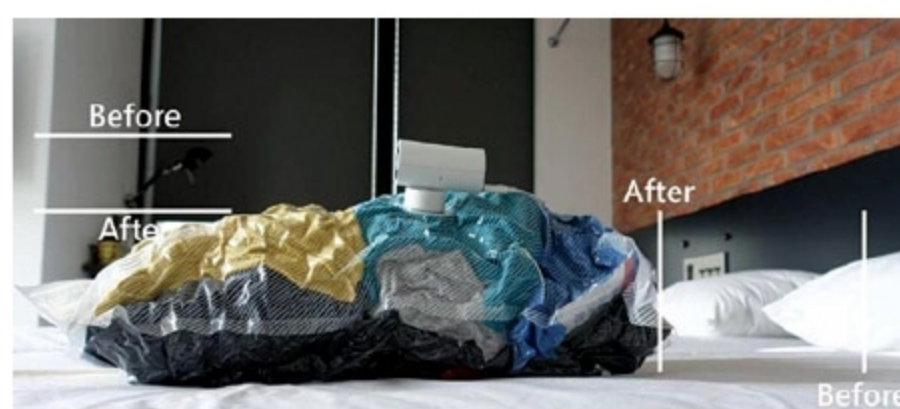
Aeronext develops and provides solutions in the industrial robot sphere based on drones that are optimal for specific applications in such industry segments as logistics and inspection.

—Deepshikha Shukla

3 A TRAVEL TOOL THAT SOLVES LUGGAGE WOES

Travelling is one of the most fun things to do as it gives us memories to cherish forever. But often, we find that luggage becomes the worst part of the process. VAGO, founded by Leon, came up with an on-the-go compression device in April 2016 to solve the luggage issues—it was a successfully-backed Kickstarter project.

About ideation, Bianca, sales manager, VAGO, says, "While we travel, we always wish to bring a lot of souvenirs back home. However, it is challenging to fit all of them in our bag due to space constraints. Our aim



is to create more space in the existing luggage to make packing easier and efficient. We designed VAGO for travellers all over the world, so they could bring more things after or before the trip."

Already-available vacuum bags give us the option of squeezing air out forcibly through a pump or vacuum cleaner. The Taiwan-based startup created multiple prototypes for over a year before coming up with the final product. Regarding the current product, Chao says, "VAGO, the small, portable vacuum device, could help users save more than fifty per cent luggage space by compress-